

Vishwas Sathish

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EDUCATION

University of Washington, Seattle

2021 - Present

PhD student in Computer Science and Engineering (Advisor: Rajesh P.N. Rao)

PES University, Bangalore

2015 - 2019

B.Tech in Computer Science and Engineering (Advisors: Rajiv Bajpai & Bhaskarjyoti Das)

WORK EXPERIENCE

Amazon (A9), Palo Alto

June 2025 - Present

Applied Scientist Intern | Multimodal models, RL finetuning, vLLM, ray, pytorch, AWS

Advisors: [Douglas Gray](#) & [Arnab Dhua](#)

Worked with the multi-modal search team at Palo Alto, CA. **Finetuned Qwen** family of models with novel RL algorithm built on **GRPO** technique for Visual Question Answering (VQA). Solved multi-hop VQA with search and tool use.

Paul G. Allen School of Computer Science and Engineering

Jan 2022 - Present

Graduate Research Assistant | Pytorch, Jax, Tensorflow, Habitat-Lab, Mujoco, Maniskill, Gymnasium

Advisor: [Rajesh P.N. Rao](#)

Focus: Sequence Models, Reinforcement Learning, Representation Learning, Vision-Language Models

- **Masked Composer:** Distilled Vision-Language Models (VLMs) to mask away task-irrelevant attributes in Visual RL. These models are highly generalizable and scale to multi-modal control tasks.
- **Active Predictive Coding:** Hierarchical world models and policies to solve multi-modal navigation tasks such as [Habitat 2.0](#). Published in Robotics Science and Systems and Neural Computation, 2024.

SensAI Healthcare (7sugar), Bangalore ([Website](#))

July 2019 - June 2021

Machine Learning Engineer | Tensorflow, Keras, Numpy, Plotly, Javascript, asyncio

Mentors: [Adhi Kesarla](#) & [Rajesh Rao](#)

Focus: Computer Vision, Hands-on Product Deployment, Autoregressive Forecasting, and Dialog Agents

- **Vision:** Deployed a real-time calorie estimator and meal classifier for more than 200 varieties of food, reducing the inference time from 5 minutes to 0.5 seconds. Received the [TIDE 2.0](#) Start-up Grant for this project.
- **Forecasting:** Trained and deployed a state-of-the-art real-time GRU-Decay RNN for auto-regressive glucose forecasting with missing data. My work was presented at the Diabetes India Conference, 2020.

Morgan Stanley, Bangalore

Jan 2019 - July 2019

Technology Analyst Intern | Data Transformation, Java, Javascript, Angular, Spring Framework

Worked on data routing and transformation using Apache Camel for Machine Learning pipelines. The software was used by multiple services within the organization, including a responsive website that I designed to search and query financial data.

Systems Biology Institute, Tokyo ([Website](#))

July 2018 - Sept 2018

Summer Research Intern | Python, Scikit-learn, Numpy, Pandas

Mentors: [Vipul Gupta](#) & [Ayako Yachi](#)

Built an [Explainable Machine Learning Framework](#) for Drug Classification and Discovery. The framework explained why a particular output class was chosen for a given set of drug compositions.

SKILLS

Tools: Pytorch, Tensorflow, Huggingface transformers, vLLM, TRL, veRL, Numpy, Scikit-Learn, Pandas, AWS, Docker, Ray, Slurm, Git, Bash, Python, C/C++, Java, Javascript, Typescript, HTML/CSS, $\text{\LaTeX}$

Expertise: Large Language Models, Multi-modal LLMs, Reinforcement Learning, RL-Finetuning, Vision Transformers, World Models, Unsupervised Representation Learning, State Space Models, Planning and Navigation

PATENTS

1. **Apparatuses, Systems, and Methods for Active Predictive Coding Networks**

2024

Full US Patent 2024 (Serial Number: 18/408,173). Patent Pending. [Google Patents Link](#)

Rajesh Rao, Dimitrios C. Gklezakos, **Vishwas Sathish**

RELEVANT PUBLICATIONS

1. **Eliciting Self-Verification in Multimodal Reasoning Agents with Reinforcement Learning**
Vishwas Sathish, Viresh Ranjan, Xinliang Zhu, Arnab Dhua, Douglas Gray.
In Submission @ ECCV 2026
2. **Masked Composer: Masked Model-Based RL for Compositional Generalization**
Vishwas Sathish, Matthew James Bryan, Sidharth, Abhishek Gupta, Rajesh P. N. Rao
In Submission
3. **Active Predictive Coding: A Unifying Neural Model for Active Perception, Compositional Learning, and Hierarchical Planning**
Rajesh Rao, Dimitrios C. Gklezakos, Vishwas Sathish.
Published @ Neural Computation, 2024 (Paper)
Presented @ Proceedings of the Annual Meeting of the Cognitive Science Society (Cogsci), 2023 (Poster)
4. **Multimodal Natural Intelligence through Active Predictive Coding**
Vishwas Sathish*, Jeffrey Duan*, Rajesh Rao
Presented @ AAAI 2026, Neuro for AI & AI for Neuro: Towards Multi-Modal Natural Intelligence (website)
5. **Learning Hierarchical Abstractions of Complex Dynamics via Prediction Errors**
Vishwas Sathish, Rajesh Rao.
Presented @ Robotics Science and Systems (RSS) 2024, Priors4Robots Workshop (workshop) (paper)
Poster @ Connectome 2024 (Poster)
6. **Decoding Pain: Statistical Identification of Biomarkers from Electrophysiological Signals**
Sidharth*, Vishwas Sathish*, Shweta Bansal, Samantha Sun, Timmy Pham, Rajesh Rao, Jefferey Herron.
Presented @ AAAI 2025, Workshop on Health Intelligence (website)

PREPRINTS AND OTHER RELEVANT PROJECTS

1. **LLeMpower: Understanding Disparities in the Control and Access of LLMs** ([arxiv preprint](#)) 2024
Vishwas Sathish, Hannah Lin, Aditya K Kamath, Anish Nyayachavadi.
2. **Lessons from Neuroscience: How integrating Compositional Structure and Episodic Memory could enable Human-Like AI** ([arxiv preprint](#)) 2024
Rajesh Rao, Vishwas Sathish, Preston Jiang, Matthew Bryan, Prashant Rangarajan
3. **Examining Large Language Model Cognition Through Puzzles and Optical Illusions.** ([Paper](#)) 2023
Vishwas Sathish, Aditya Kamath, Raphael Bechtold
4. **Graph Embedding Based Hybrid Social Recommendation System** ([Paper](#)) 2019
Vishwas Sathish, Tanya Mehrotra, Simran Dhinwa, Bhaskarjyoti Das
5. **Sentiment Analysis for Product Reviews with GloVE embeddings and LSTMs** ([Github](#)) 2017

PROFESSIONAL RESPONSIBILITIES

- **Academic Review Service** - ICML 2026, ICLR 2025-2026, Neurips 2025, ICCV 2025, RSS 2024 2024 - 2026
- **Graduate Application Reader** (ML/Neural Computing Track), Paul G. Allen School of CSE 2022 - 2025
- **Student Area Chair** (Neural Computing Track): PhD Admissions, Paul G. Allen School of CSE 2022
- **Pre-Application Mentorship Service, Mentor** ([PAMS](#)), Paul G. Allen School of CSE 2022
- **Graduate Teaching Assistant** ([UW CSE 421](#)), Paul G. Allen School of CSE 2021
- **Mentoring** - Sidharth (CS PhD, UMichigan), Jeffrey Duan (Incoming CS PhD @ UW), Yubo Zhang (PhD @ UW)

AWARDS AND ACHIEVEMENTS

- **ACM ICPC**: Bagged Rank 1 at the institution level ACM-ICPC and represented PES University at the regional contest in Chennai. Two-time top 10 rank in ICPC Pacific NW Qualifiers. 2018, 2022, 2023
- **TIDE 2.0 Startup Grant**: [Received the grant](#) for my work on meal classification at SensAI Healthcare. 2019
- 2 time recipient of **CNR Rao Metrit Scholarship** awarded by PES University for academic excellence. 2017, 2018
- **SOF Science Olympiad**: Gold Medalist with a state rank of 14 and international rank 410. 2015
- Among **Top 0.4%** of 150,000 candidates appearing in the 2015 KCET Examinations. 2015
- Among **Top 1%** of 1,300,000 students appearing in the 2015 JEE Mains Examinations. 2015